16 n is an integer greater than 1

Prove algebraically that $n^2 - 2 - (n - 2)^2$ is always an even number.

×	n	-2
n	n^2	$-2n$
-2	$-2n$	$+4$

$(n - 2)^2 = n^2 - 2n - 2n + 4$

1 mark
Expand the bracket correctly

$n^2 - 2 - (n^2 - 4n + 4)$

$= n^2 - 2 - n^2 + 4n - 4$

1 mark
Subtract every term in the bracket

$= 4n - 6 = 2(2n - 3)$

1 mark
Collect and factor out 2

$2n - 3$ is an integer, so $2(2n - 3)$ is always even.

1 mark
A multiple of 2 for integer n

(Total for Question 16 is 4 marks)